CS23334 Fundamentals of Data Science Lab

Experiment 15

Data Preprocessing Pipeline & Encoding in Data Science

Objective: To understand the end-to-end data preprocessing pipeline including missing value handling (mode, median, and mean imputation), categorical dummy encoding (`pd.get_dummies`), and binary target encoding.

Methodology:

1. **Initial Inspection & Missing Value Audit:** Load dataset (`pre-process_datasample.csv`) and inspect missing attributes across numerical and categorical features using `df.info()`.
2. **Multi-type Imputation Strategy:** Impute missing categorical attributes using feature mode (`Country.mode()[0]`), missing discrete attributes using feature median (`Age.median()`), and missing continuous attributes using feature mean (`Salary.mean()`).
3. **One-Hot Dummy Encoding:** Generate binary boolean indicators for categorical variables using `pd.get_dummies(df.Country)` and concatenate with numerical feature subsets using `pd.concat(..., axis=1)`.
4. **Target Encoding:** Transform binary target strings (`No`, `Yes`) into numeric integers (0, 1) using `.replace()`.

Code:

```
import numpy as np
import pandas as pd

df = pd.read_csv("pre-process_datasample.csv")
df

df.info()

df.Country.mode()
df.Country.mode()[0]
type(df.Country.mode())

df.Country.fillna(df.Country.mode()[0], inplace=True)
df.Age.fillna(df.Age.median(), inplace=True)
df.Salary.fillna(round(df.Salary.mean()), inplace=True)

df
```

```
pd.get_dummies(df.Country)

updated_dataset = pd.concat([pd.get_dummies(df.Country), df.iloc[:, [1, 2, 3]]], axis=1)
updated_dataset

df.info()

updated_dataset.Purchased.replace(['No', 'Yes'], [0, 1], inplace=True)
updated_dataset
```

OUTPUT: 1. Initial Dataset, Mode Inspection & Imputed DataFrame

```
In [1]: df = pd.read_csv("pre-process_datasample.csv")
In [2]: df
Out[2]:
   Country   Age   Salary Purchased
0   France  44.0  72000.0        No
1    Spain  27.0  48000.0       Yes
2  Germany  30.0  54000.0        No
3    Spain  38.0  61000.0        No
4  Germany  40.0      NaN       Yes
5   France  35.0  58000.0       Yes
6    Spain   NaN  52000.0        No
7   France  48.0  79000.0       Yes
8      NaN  50.0  83000.0        No
9   France  37.0  67000.0       Yes

In [3]: df.info()
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 4 columns):
 #   Column     Non-Null Count  Dtype
---  ------     --------------  -----
 0   Country    9 non-null      object
 1   Age        9 non-null      float64
 2   Salary     9 non-null      float64
 3   Purchased  10 non-null     object

In [4]: df.Country.mode()[0]
Out[4]: 'France'

In [5]: df.Country.fillna(df.Country.mode()[0], inplace=True)
In [6]: df.Age.fillna(df.Age.median(), inplace=True)
In [7]: df.Salary.fillna(round(df.Salary.mean()), inplace=True)
In [8]: df
Out[8]:
   Country   Age   Salary Purchased
0   France  44.0  72000.0        No
1    Spain  27.0  48000.0       Yes
2  Germany  30.0  54000.0        No
3    Spain  38.0  61000.0        No
4  Germany  40.0  63778.0       Yes
5   France  35.0  58000.0       Yes
6    Spain  38.0  52000.0        No
7   France  48.0  79000.0       Yes
8   France  50.0  83000.0        No
9   France  37.0  67000.0       Yes
```

OUTPUT: 2. Dummy Feature Encoding & Final Target Encoded Dataset

```
In [9]: pd.get_dummies(df.Country)
Out[9]:
   France  Germany  Spain
0    True    False  False
1   False    False   True
2   False     True  False
3   False    False   True
4   False     True  False
5    True    False  False
6   False    False   True
7    True    False  False
8    True    False  False
9    True    False  False

In [10]: updated_dataset = pd.concat([pd.get_dummies(df.Country), df.iloc[:, [1, 2, 3]]], axis=1)
In [11]: updated_dataset.Purchased.replace(['No', 'Yes'], [0, 1], inplace=True)
In [12]: updated_dataset
Out[12]:
   France  Germany  Spain   Age   Salary  Purchased
0    True    False  False  44.0  72000.0          0
1   False    False   True  27.0  48000.0          1
2   False     True  False  30.0  54000.0          0
3   False    False   True  38.0  61000.0          0
4   False     True  False  40.0  63778.0          1
5    True    False  False  35.0  58000.0          1
6   False    False   True  38.0  52000.0          0
7    True    False  False  48.0  79000.0          1
8    True    False  False  50.0  83000.0          0
9    True    False  False  37.0  67000.0          1
```

Result:

A complete end-to-end preprocessing pipeline was successfully implemented. Missing values were imputed using mode, median, and mean for categorical, discrete, and continuous features respectively. One-hot encoding via pd.get_dummies produced a fully numeric feature matrix ready for model training.